Sesiune - FAI - SESIUNE

Considerăm următoarea procedură de codificare a unui șir de caractere s, fără diacritice și fără distincție între litere mici și mari:

1. Se convertește fiecare caracter al șirului s într-un întreg folosind tabelul din Figura 1;

2. Se face suma modulo 26 a întregilor rezultați;

3. Dacă întregul de la pasul anterior este $\neq 0$ atunci el este rezultatul final al procedurii; altfel se consideră 1 ca fiind rezultatul final al procedurii.

Vom nota rezultatul prin <s'>.

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

t	u	v	w	x	y	z
19	20	21	22	23	24	25

Subiectul nr 1 al examenului

Folosind AEE, determinați cel puțin o soluție a ecuației

<prenume>x + <nume>y = c.

<prenume> = <DIANA> = (3+8+0+13+0) % 26 = (16+8) % 26 = 24 % 26 = 24.

(întregul rezultat este $\neq 0 \Rightarrow$ el este rezultatul procedurii).

<nume> = <UNGUREANU> = (20+13+6+20+17+4+0+13+20) % 26

= 113 % 26 = 9.

Așadar, putem forma ecuația $24x + 9y = c$.

Pași:

$\exists$ cel puțin o soluție $\Leftrightarrow (a,b) \mid c \Leftrightarrow (24,9) \mid c \Leftrightarrow 3 \mid c \Rightarrow c = 3$.

Pas 2: Aplicăm AEE: Fie $V_e = (\alpha, \beta)$ a.î. $\alpha \cdot a + \beta \cdot b = (a,b) \Leftrightarrow \alpha \cdot 24 + \beta$

Împărțiri

$24 = 2 \cdot 9 + 6$

$9 = 1 \cdot 6 + 3$

$6 = 2 \cdot 3 + 0$

(am terminat)

AEE

$V_{24} = (1,0)$; $V_9 = (0,1)$;

$V_6 = V_{24} - 2 V_9 = (1,0) - 2 \cdot (0,1)$

$= (1,-2)$.

$V_3 = V_9 - V_6 = (0,1) + (-1,2)$

$= (-1,3)$.

$\Rightarrow$ verificare: $24 \cdot (-1) + 9 \cdot 3 = 27 - 24 = 3$ (A).

Soluția ecuației prin AEE este $(-1,3)$.

Exercițiul 2

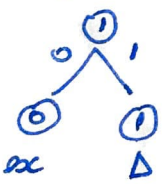
prenume // nume

DIANA // UNGUREANU

DUINAGNUAR (șir de 10 caractere)

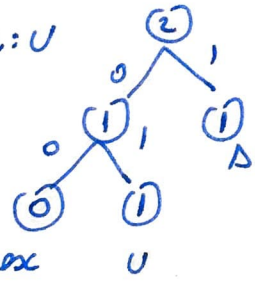
Aplicăm Huffman adaptiv.

A1: D



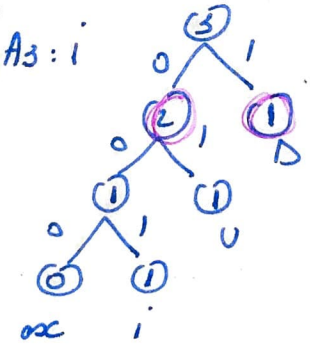
sibling: 011

A2: U



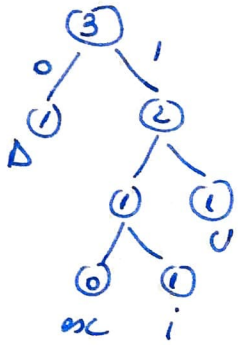
sibling: 01112

A3: I



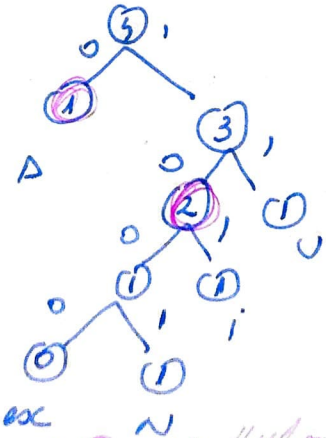
sibling: 0111213

$\rightsquigarrow$



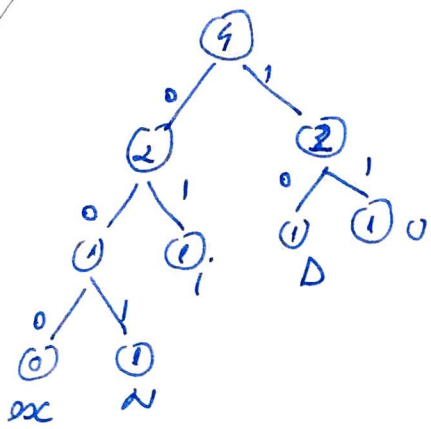
sibling: 0111123

A4: N

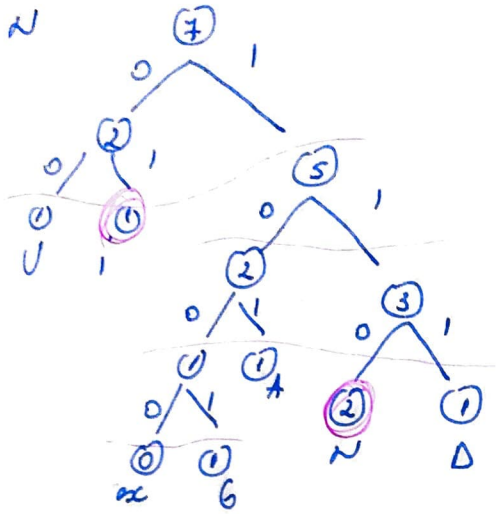


sibling: 01112134 // nul mai din st. cu cel mai din dr.

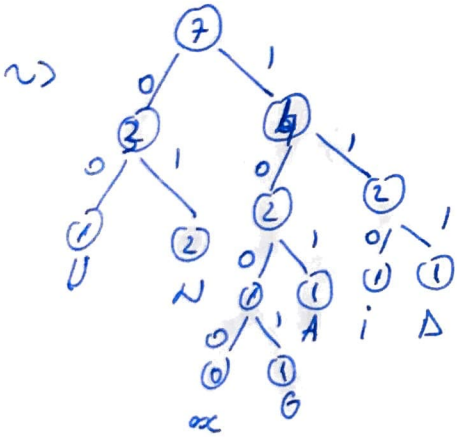
A4: N


4
2
2
1
1
1
1
1
0
1
0
1
0

A7: N

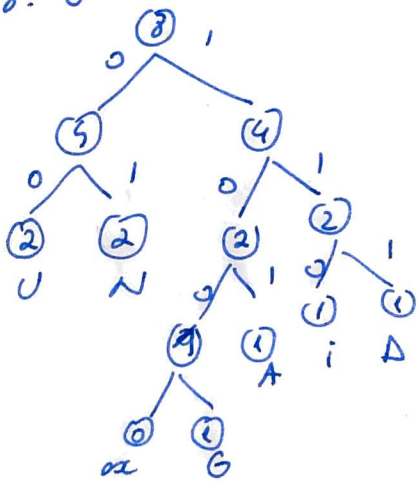


sibling: 0111211123257



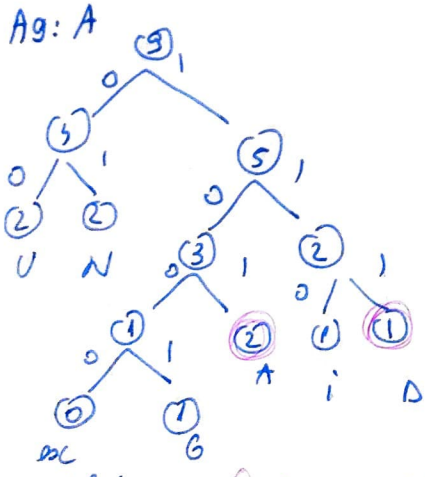
sibling: 011111122234

A8: U

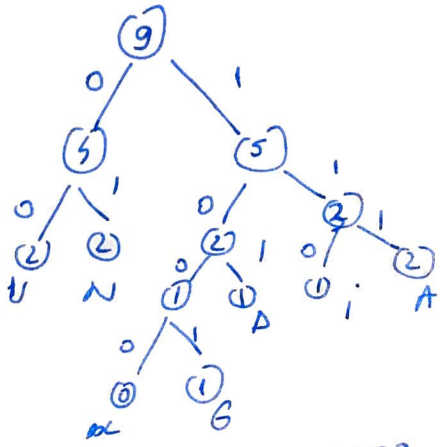


sibling: 011112222448

A9: A

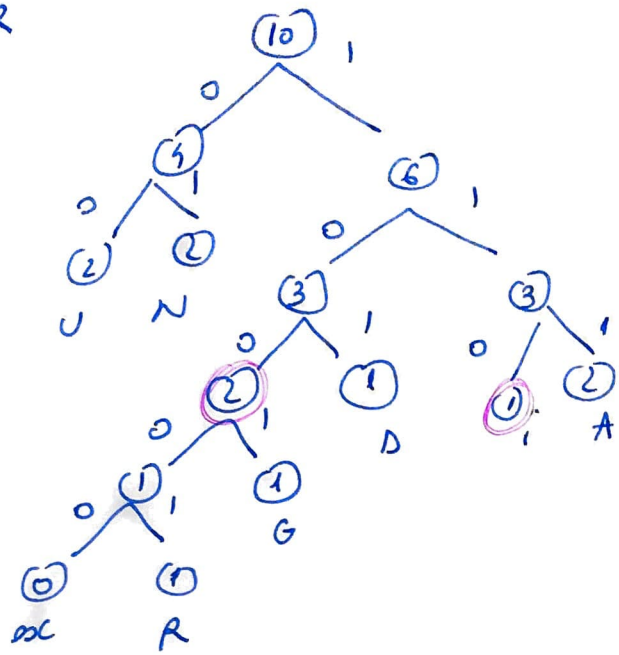


sibling: 0112112232459



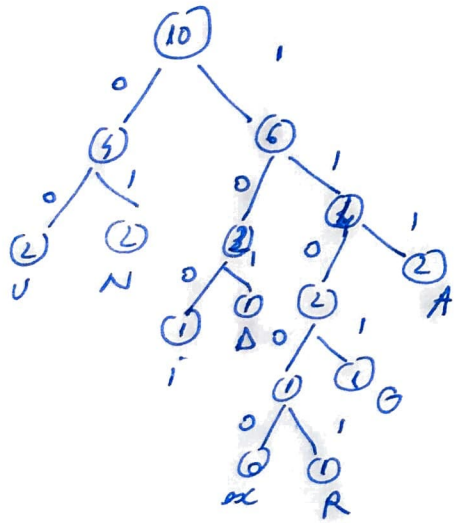
sibling: 011112222459

o: R



sibling: 0 1 1 1 2 1 1 2 2 2 3 3 4 6 1 0

~>



sibling: 0 1 1 1 1 1 2 2 2 2 4 4 6 1 0 (rezultatul final)